START HERE

# Tape Diagrams

A step-by-step guide. Read one page at a time. Look at the pictures. Then try the **Your turn** boxes.




**IN THIS BOOK YOU WILL LEARN**

1. What a tape diagram is
2. Long tape, short tape
3. Parts and the whole
4. The hiding number ?
5. Read · Draw · Write
6. Adding stories
7. Missing part stories
8. Take-away stories
9. Comparing stories
10. Equal parts
11. Golden rules
12. Practice + answers

**My name:** _____ **Class:** _____

# What is a tape diagram?

A tape diagram is a **long box** that shows a number. It looks like a strip of tape. It helps us **see** a story.

1. **Aman has 3 mangoes.**



2. **Put each mango in a box.**



3. **Push the boxes together. No gaps!**



It looks like a strip of tape.

4. **Drawing mangoes is slow. Just write the number 3.**




The **number** goes inside. The **label** (mangoes) tells what it is.

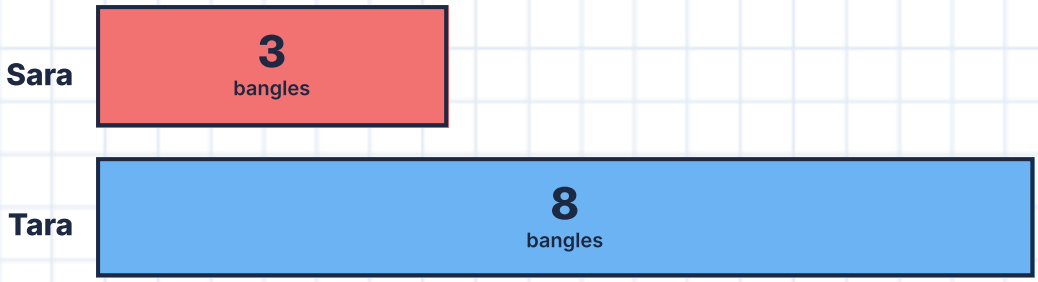
**REMEMBER**

**50 mangoes?** Do not draw 50 mangoes! Draw one tape and write 50. Tape diagrams are quick.



# Long tape or short tape?

Sara has **3** bangles. Tara has **8** bangles.




Tara's tape is longer, because 8 is bigger than 3.

REMEMBER

## Bigger number → longer tape

Smaller number → shorter tape. Same number → same length.

You do not need a ruler. The tape only needs to **look** about right.

YOUR TURN

1. Which tape should be longer, 7 or 4? _____
2. Which tape should be shorter, 10 or 2? _____
3. Draw a tape for 2 and a tape for 6 below.

Draw here

# Parts and the whole

Pinky has 4 red balloons and 2 blue balloons.



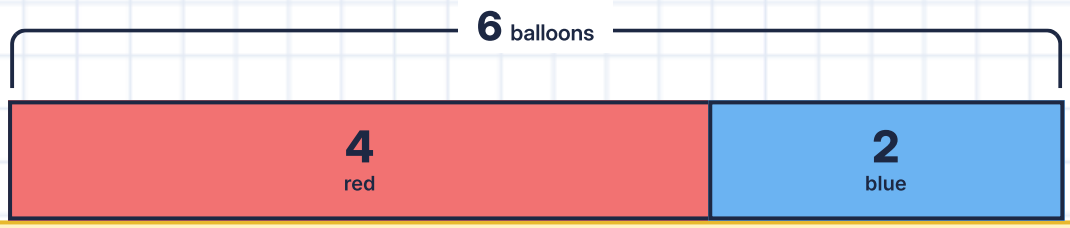
**1** Red balloons are one PART.




**2** Blue balloons are another PART. Stick it next to the red part.




**3** All the balloons together is the WHOLE. Show it with a bracket on top.




MAGIC SENTENCE

## PART + PART = WHOLE

4 + 2 = 6

YOUR TURN

Look at this tape. Write the parts and the whole.

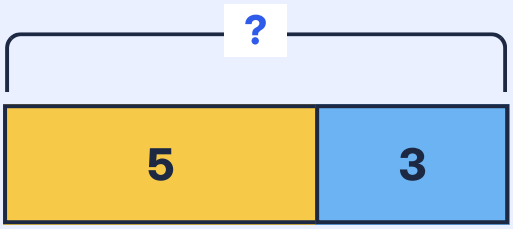



Parts: _____ and _____ Whole: _____

# The hiding number ?

In every story, one number is **hiding**. We write a **?** for it. The ? says: **"Find me!"**

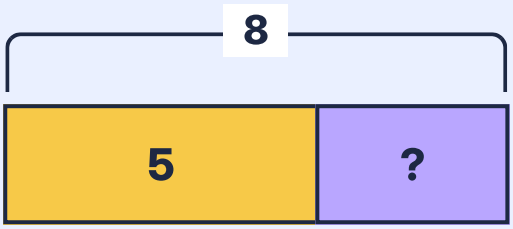
### TAPE A: THE WHOLE IS HIDING




We know both parts. We need the whole.

**ADD: 5 + 3 = 8**

### TAPE B: A PART IS HIDING




We know the whole and one part.

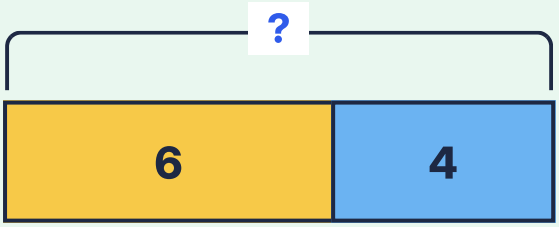
**SUBTRACT: 8 − 5 = 3**

### SUPER SECRET TRICK

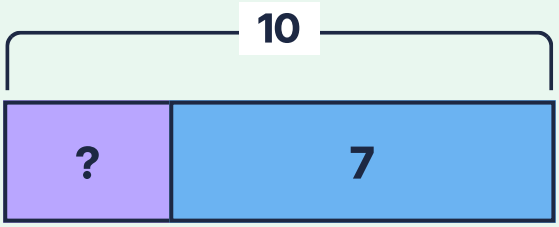
| What is hiding? | What do I do? |
|---|---|
| The **WHOLE** (? on top) | **ADD +** |
| A **PART** (? in a box) | **SUBTRACT −** |

### YOUR TURN

What is hiding? Write **part** or **whole**. Then write + or −.




Hiding: _____ Sign: _____




Hiding: _____ Sign: _____

STEP 5

# Three steps: Read, Draw, Write

Every story problem, every time:

1. **READ the story two times.**
   - **Who** is it about?
   - **What** things?
   - Which **numbers** do I see?
   - What is the **question** asking?
2. **DRAW the tape.**
   - Draw a box for each part. Bigger number → longer box.
   - Write the numbers inside.
   - Put a **?** where the number is hiding.
   - Write labels (names of things).
3. **WRITE.**
   - A number sentence, like **5 + 3 = 8**.
   - An answer sentence, like **Riya has 8 pencils.**

REMEMBER

## READ · DRAW · WRITE

Use actions: open your hands like a book (READ), draw a box in the air (DRAW), write in the air (WRITE).

# Riya's pencils

Riya has 5 pencils. Her mom gives her 3 more pencils. How many pencils does Riya have now?

1. **READ.**
   Who? **Riya**. What? **pencils**. Numbers? **5** and **3**. Question: how many **now**?

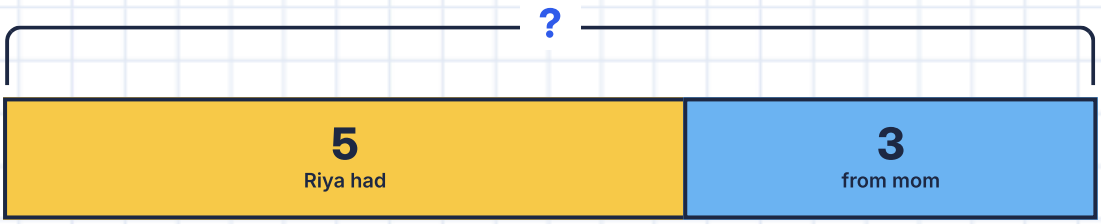
2. **DRAW the first part: 5.**




3. **"3 more" means add another part. 3 is smaller, so a shorter box.**




4. **The question asks for all the pencils: the WHOLE. Put ? on top.**




5. **The whole is hiding → ADD. WRITE:**

**5 + 3 = 8**

**Riya has 8 pencils now.**

EXAMPLE 2 · MISSING PART

# Birds on a tree

There are 12 birds on a tree. 7 birds are brown. The rest are white. How many birds are white?

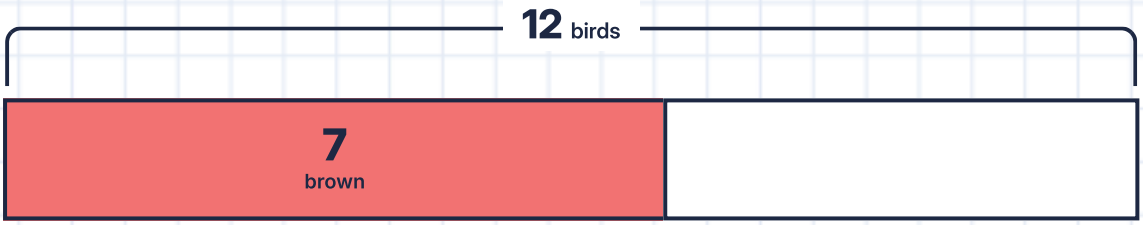
**1** **READ.**

Numbers: 12 and 7. **12 is ALL the birds**, so 12 is the **whole**.

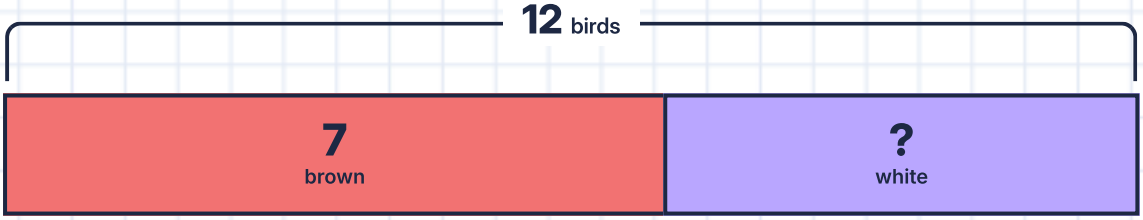
**2** **DRAW the whole tape first. Write 12 on the bracket.**




**3** **7 are brown. That is one part.**




**4** **The rest are white. We do not know how many → ?**




**5** **A part is hiding → SUBTRACT. WRITE:**

**12 − 7 = 5**

**5 birds are white.**

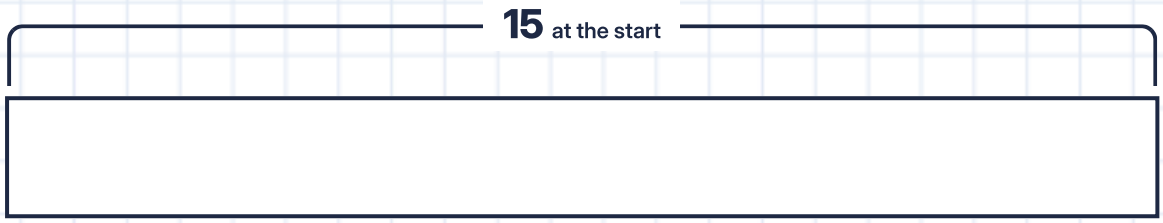
Check: 7 + 5 = 12 ✔ (part + part = whole)

EXAMPLE 3 · TAKING AWAY

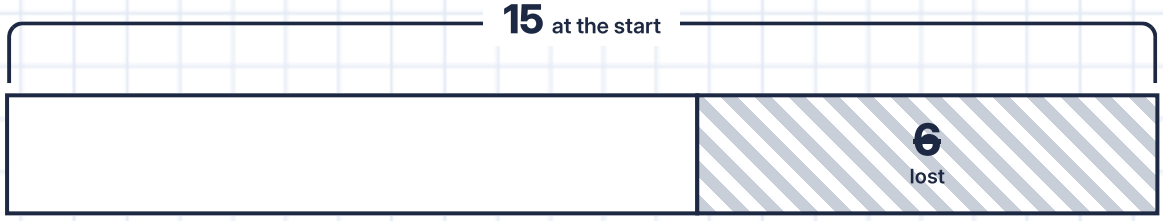
# Kabir's lost marbles

Kabir had 15 marbles. He lost 6 marbles. How many marbles does he have left?

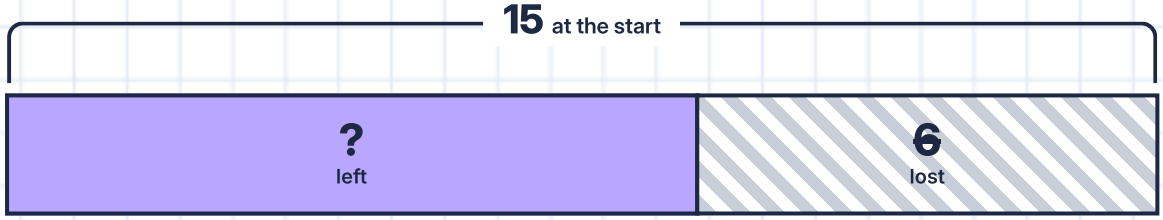
**1** **Kabir HAD 15 at the start. That is the whole.**




**2** **He LOST 6. Mark a part of 6 and cross it out.**




**3** **The marbles left are hiding → ?**




**4** **A part is hiding → SUBTRACT. WRITE:**

**15 − 6 = 9**

**Kabir has 9 marbles left.**

Check: 9 + 6 = 15 ✔

**TAKE-AWAY CLUE WORDS**

**lost · ate · gave away · flew away · broke · spent · left**

These words often mean a part was taken away.

EXAMPLE 4 · COMPARING

# Meera and Dev's stickers

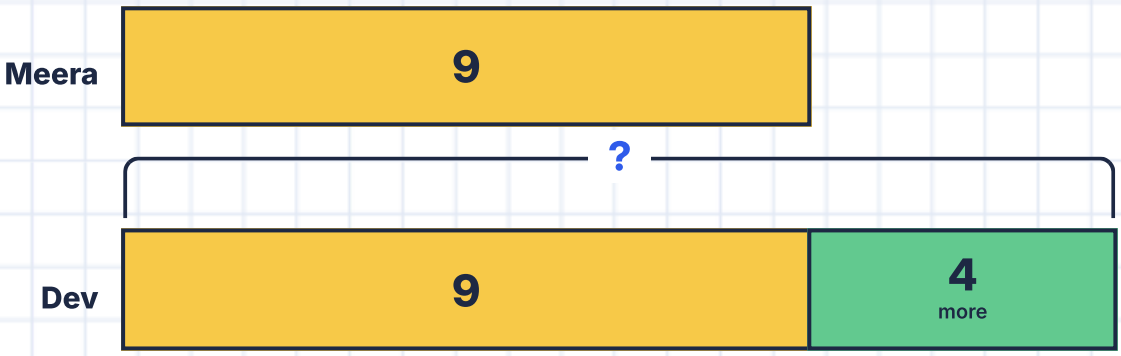
Meera has 9 stickers. Dev has 4 more stickers than Meera. How many stickers does Dev have?

1. **Two people → draw TWO tapes, one under the other.**

2. **Meera's tape: 9.**




3. **Dev has the SAME as Meera, AND 4 more. Start Dev's tape exactly under Meera's.**




4. **Dev's whole tape is hiding → ADD. WRITE:**

**9 + 4 = 13**

**Dev has 13 stickers.**

**REMEMBER**

**Line up the left edges** of both tapes. The extra piece shows **how many more**. Here: Dev has **4 more**.

EXAMPLE 5 · EQUAL PARTS

# Boxes of laddoos

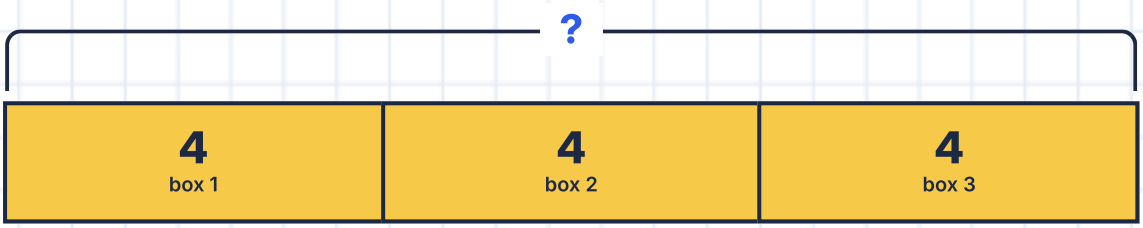
There are 3 boxes. Each box has 4 laddoos. How many laddoos are there in all?

**1** **3 boxes → 3 parts. Each box has 4 → write 4 in every part.**




**2** **Same number → same size. These are EQUAL parts.**

**3** **All the laddoos is the whole → ?**




**4** **ADD all the parts. WRITE:**

**4 + 4 + 4 = 12**

**There are 12 laddoos in all.**

YOUR TURN

What if there were **5** boxes with 4 laddoos each? Draw the tape and find the whole.

Draw here

Number sentence: _____

REMEMBER

# Tape diagram golden rules

1. **Read the story two times.**
2. **Bigger number → longer tape.**
3. **Part + Part = Whole.**
4. **Put a ? on the hiding number.**
5. **Whole hiding → ADD. Part hiding → SUBTRACT.**
6. **Write labels. Answer in a full sentence.**

### WHICH TAPE DO I DRAW?

| The story says... | Draw this |
|---|---|
| Two amounts put together (and, more, in all) | One tape, two parts, **?** on top |
| All of them, and some are one kind | Whole on top, one part known, **?** part |
| Had some, then lost / ate / gave away | Whole on top, cross out a part, **?** left |
| Two people, "more than" / "fewer than" | Two tapes, left edges lined up |
| Groups with the same number in each | Equal parts |

PRACTICE

# Your turn! (1)

Draw the tape. Put a **?** on the hiding number. Then write.

**PROBLEM 1**

**Ali has 6 red cars and 5 blue cars. How many cars does he have in all?**

Draw here

Number sentence: _____ Answer: _____

**PROBLEM 2**

**There are 14 children in a class. 8 are girls. How many are boys?**

Draw here

Number sentence: _____ Answer: _____

**PROBLEM 3**

**Sia had 10 balloons. 3 balloons flew away. How many balloons are left?**

Draw here

Number sentence: _____ Answer: _____

PRACTICE

# Your turn! (2)

**PROBLEM 4**

A bus has 9 people. 7 more people get on. How many people are on the bus now?

Draw here

Number sentence: _____ Answer: _____

**PROBLEM 5**

Raj has 13 shells. Tina has 5 more shells than Raj. How many shells does Tina have? (Two tapes!)

Draw here

Number sentence: _____ Answer: _____

**PROBLEM 6**

There are 4 plates. Each plate has 3 idlis. How many idlis are there in all? (Equal parts!)

Draw here

Number sentence: _____ Answer: _____

# Answers

Check your work. If your tape looks a little different but the numbers and the ? are in the right places, it is correct!

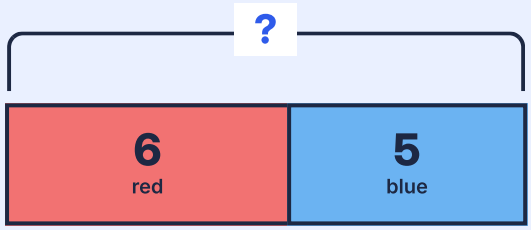
**Step 2:** 1. 7  2. 2

**Step 3:** Parts 3 and 5 · Whole 8

**Step 4:** Left: whole, + (6 + 4 = 10) · Right: part, − (10 − 7 = 3)

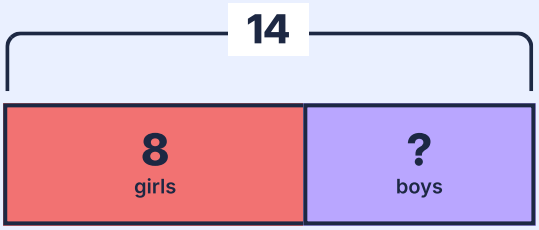
**Step 5 (5 boxes):** 4 + 4 + 4 + 4 + 4 = 20

### 1. ALI



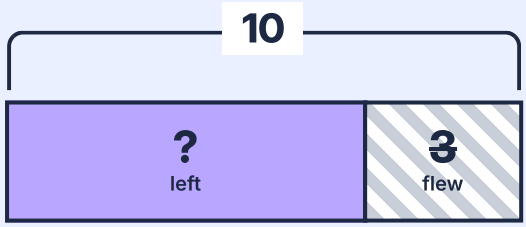

6 + 5 = 11 · Ali has 11 cars.

### 2. CLASS



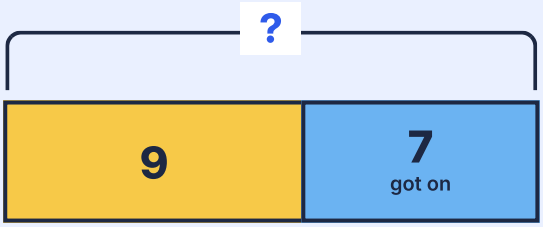

14 − 8 = 6 · There are 6 boys.

### 3. SIA



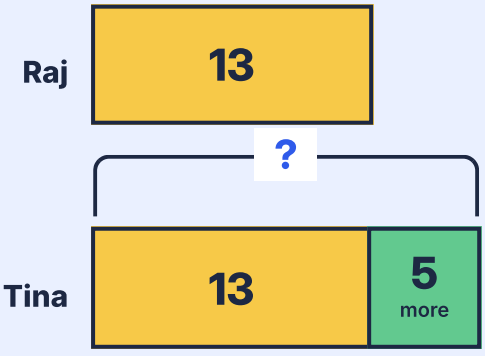

10 − 3 = 7 · Sia has 7 balloons left.

### 4. BUS




9 + 7 = 16 · 16 people.

### 5. RAJ AND TINA




13 + 5 = 18 · Tina has 18 shells.

### 6. IDLIS




3 + 3 + 3 + 3 = 12 · 12 idlis.